

Classifying Eye State from EEG Signals via k -Nearest Neighbours: A Cross-Validation Honesty Study

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Abstract

We pick a single data-analysis approach from the COGS 109 study guide—**classification**—and a single model within that approach— **k -nearest neighbours (KNN)**—and we perform model selection by sweeping the hyperparameter k over a log-spaced grid $\{1, 3, 5, 7, 11, 15, 21, 31, 51, 75, 99, 151, 201\}$ to find the value that minimises 5-fold cross-validation (CV) error on the UCI #264 EEG Eye State dataset (Roesler, 2013). We then run that *same* model-selection procedure under three different cross-validation schemes: shuffled, naive blocked, and stratified blocked. The contribution is *methodological*: we document how the choice of CV scheme changes the selected hyperparameter and the reported accuracy on an autocorrelated EEG recording. All three schemes select $k = 1$, but the reported accuracies differ dramatically: $97.3\% \pm 0.4\%$ under shuffled CV, $50.0\% \pm 10.7\%$ under naive blocked CV, and $77.8\% \pm 2.7\%$ under honest stratified blocked CV. The 19.5 *pp* gap between the shuffled and stratified-blocked estimates is leakage attributable to scheme choice—not to model quality, since the identical KNN model is evaluated in all three settings on the same training data. A sanity-check appendix runs the same idea on three alternative classifiers (LDA, PCA→LDA, PCR-as-classifier) and finds leakage gaps of only 2–6 *pp*, consistent with KNN being uniquely vulnerable to lag-1 autocorrelation because it predicts from per-sample similarity.

Shuffled CV says we have a 97% classifier. Honest stratified blocked CV says we have a 78% classifier. The difference is leakage, not signal.

1 Introduction

1.1 Problem and motivation

Electroencephalography (EEG) eye-state classification—deciding from raw scalp voltages whether a subject’s eyes are open or closed—is a building block of brain–computer interfaces, drowsiness detection, and attention-monitoring systems. The UCI #264 EEG Eye State dataset is one of the most widely used benchmarks for this task, and online tutorials and student write-ups routinely report k -nearest-neighbour accuracies in the 95–98% range on it under standard shuffled k -fold cross-validation.

We argue in this report that those numbers reflect a *methodological artefact* rather than a strong classifier. The dataset is a single ~ 117 -second continuous recording from one subject; its label changes only 23 times across 14,980 samples; adjacent samples are nearly identical (lag-1 channel correlation ≈ 0.997); and the label is piecewise-constant over runs that are sometimes thousands of samples long. A KNN classifier with $k = 1$ evaluated under shuffled k -fold CV is essentially asking “what was the eye state of the sample 8 ms before or after this one?”—a question to which it can almost always answer correctly. That is a property of the cross-validation rule, not of the classifier.

The question this report addresses is therefore methodological rather than predictive:

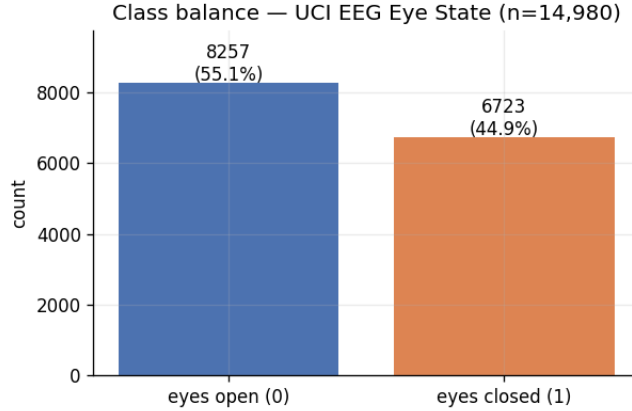


Figure 1: Class balance of the UCI #264 EEG eye-state dataset. The majority class (eyes open) sets the random-guess baseline at 0.5512.

Given an autocorrelated single-subject EEG recording and a KNN classifier, how does the choice of cross-validation scheme change the result of an otherwise identical model-selection procedure?

This matters because cross-validation is the standard tool students are taught for honest accuracy estimation, yet on autocorrelated time-series data the default i.i.d. shuffled scheme silently inflates the estimate. Quantifying that inflation on a clean, well-known benchmark makes the failure mode concrete and reproducible.

1.2 Dataset and key characteristics

We use the UCI EEG Eye State dataset (Roesler, 2013; UCI Machine Learning Repository #264): a single uninterrupted recording from one subject wearing an Emotiv EPOC consumer-grade EEG headset.

- **Observations:** $n = 14,980$ samples, recorded at ~ 128 Hz over ~ 117 s.
- **Predictors:** $p = 14$ continuous voltage channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4).
- **Response:** the binary label `eyeDetection` (0 = eyes open, 1 = eyes closed), annotated by the experimenter.
- **Class balance:** 55.12% open (8,257 samples) vs. 44.88% closed (6,723 samples), so the majority-class baseline—the accuracy floor any useful classifier must beat—is 0.5512 (Figure 1).
- **Noise sources:** consumer-grade hardware prone to electrode drift, motion and muscle artefacts, and occasional headset disconnections (visible as a handful of extreme-voltage outliers).

A purely structural feature of this dataset is the heart of the study. The labels change rarely: the recording contains only **24 contiguous label runs** (the subject opens or closes their eyes only 23 times in ~ 117 s), with run lengths ranging from under a hundred to several thousand consecutive samples (Figure 2). Consequently the lag-1 autocorrelation of any single voltage channel is ≈ 0.997 —adjacent samples are nearly identical—and the label, being piecewise-constant over long runs, has very high short-lag autocorrelation: ≈ 1.0 at lag 1, ≈ 0.83 at lag 50 (≈ 0.4 s), crossing zero only at very long lags (Figure 3).

This is the structural problem at the heart of the report: *any* CV protocol that splits samples uniformly at random leaves each test sample with a near-duplicate in its own training set. A model whose prediction depends on per-sample similarity—KNN with $k = 1$ being the canonical example—will appear spectacular, but the score reflects memorisation, not a generalisable voltage-to-label rule.

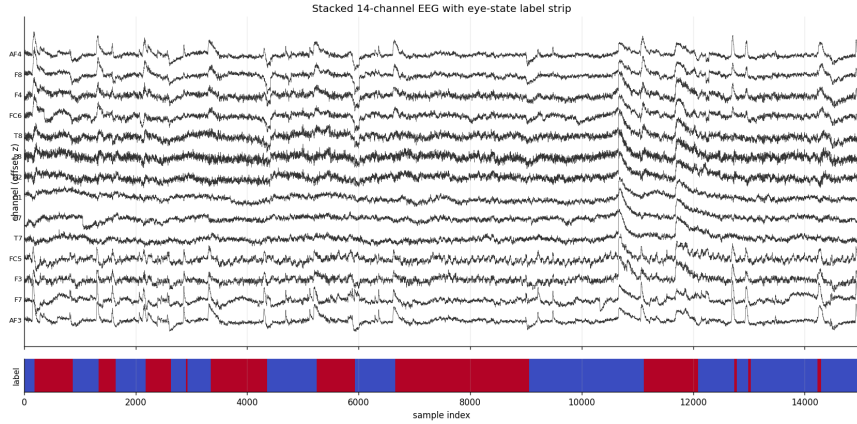


Figure 2: Raw 14-channel EEG voltages (top) with the eye-state label strip (bottom). Channels are continuous and smooth on the millisecond scale; the label is piecewise-constant over multi-second runs.

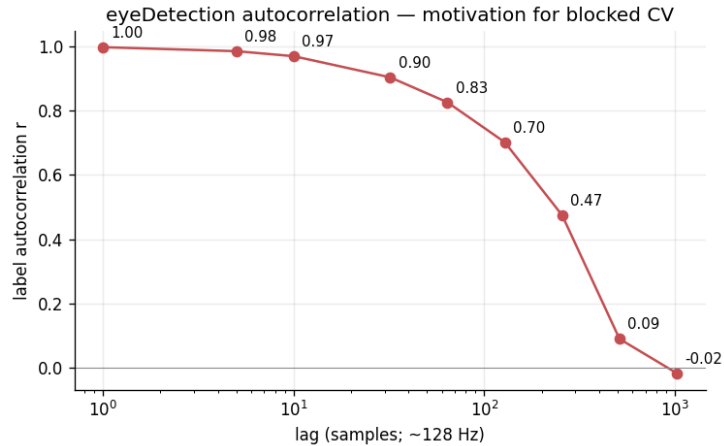


Figure 3: Autocorrelation of the binary `eyeDetection` label across lags 1–1000. The label is essentially identical to itself for the first ~ 100 samples (~ 0.8 s) and only decorrelates at very long lags. This is the structural reason shuffled CV leaks: a sample’s nearest neighbour in time is almost always in its own class.

1.3 Hypothesis

Hypothesis. A KNN classifier trained on raw 14-channel EEG voltages can predict per-sample eye state above the 55.12% majority-class baseline, but accuracy estimates obtained under shuffled k -fold cross-validation will substantially exceed honest out-of-time estimates because of temporal leakage between adjacent samples.

The hypothesis has two parts and we test both: (i) the honest accuracy beats the majority baseline, and (ii) the shuffled estimate is inflated relative to a leakage-resistant estimate. A negative or ambiguous result on either part would still be informative; we report exactly what we find.

2 Methods

2.1 Approach: classification (prediction)

The COGS 109 study guide enumerates several broad approaches (linear, polynomial and spline regression, PCR, LDA, KNN, K -means, hierarchical clustering, PCA). The target variable here is the binary `eyeDetection` label, so **classification** is the natural approach: we want a model that maps a 14-dimensional voltage vector to one of two discrete labels. Regression approaches do not target a

discrete label without artificial thresholding, and clustering approaches ignore the label entirely. We are focused on **prediction** (out-of-sample accuracy of the eye-state label), not on inference about which channel “causes” an eye state; the methodological question is specifically about how honestly we can *estimate* that predictive accuracy.

2.2 Model: k -nearest neighbours

Within classification we chose KNN as our model. KNN assigns a test sample the majority label among its k nearest training samples under Euclidean distance in the z -scored 14-D channel space. We chose it for three reasons. First, it is the simplest non-parametric classifier in the palette: it makes no distributional assumption and has exactly one hyperparameter (k) controlling the bias–variance trade-off, making it ideal for a clean single-axis hyperparameter sweep. Second—and most importantly for this study—because KNN predicts from per-sample similarity to the training set, it is the model in the palette that *most directly* exploits the lag-1 autocorrelation of EEG data; if we want to study how CV scheme choice changes a result, picking the model most sensitive to that choice gives the cleanest empirical signal. Third, KNN at small k reproduces the $\sim 97\%$ literature numbers on this dataset under shuffled CV, so the leakage gap we measure is a like-for-like comparison between an honest and a leaky estimate of the same baseline.

2.3 Preprocessing

1. **Outlier removal.** We drop the four samples with at least one channel exceeding 4 standard deviations above the channel mean ($z > 4$). This is standard EEG hygiene for headset disconnections and artefacts; an ablation (`tables/02_outlier_ablation.csv`) shows it does not materially change downstream accuracy but stabilises per-channel summary statistics. Cleaning leaves 14,976 samples.
2. **Chronological split.** We split the cleaned recording chronologically into 80% train ($n = 11,917$) and 20% test ($n = 2,995$), inserting a **64-sample seam gap** (0.5 s at 128 Hz) between train and test so no train sample is within half a second of a test sample. This is the minimum cross-partition isolation warranted by the autocorrelation profile in Figure 3.
3. **z -scoring (train statistics only).** We z -score each channel using the *training* mean and standard deviation only, and reuse those parameters for the test set. z -scoring before the split would itself leak (the test set would shape the channel scale), so we defer it to after the split.

2.4 Model-selection procedure: the k -sweep

The hyperparameter to tune for KNN is k , the number of neighbours that vote. We sweep

$$k \in \{1, 3, 5, 7, 11, 15, 21, 31, 51, 75, 99, 151, 201\},$$

a log-spaced, odd-only grid (odd avoids ties on a binary label) that starts at $k = 1$ (the most leakage-prone setting) and extends to $k = 201$ (a local average that should damp out lag-1 similarity if a real classification signal exists). For each k and each CV scheme we run 5-fold cross-validation on the z -scored training partition and aggregate per-fold accuracy as mean $\pm$ standard deviation. The **selected** k under a scheme is the argmax of mean CV accuracy. Crucially, the selection *rule* (argmax accuracy) is identical across schemes—only the fold-assignment rule changes.

2.5 Cross-validation schemes (how the splitting actually works)

We implement all three schemes ourselves rather than delegating to a library, so we can state exactly how each one partitions the $N = 11,917$ training samples into 5 folds. In every scheme each sample appears in exactly one test fold, and a fold’s training set is the complement of its test set.

Scheme A — shuffled 5-fold (leaky baseline). We generate an index array $[0, 1, \dots, N - 1]$ and apply a single seed-fixed Fisher–Yates permutation (NumPy `default_rng(42)`). The shuffled indices are cut into 5 contiguous equal blocks (the last block absorbs the remainder); block i is the test set

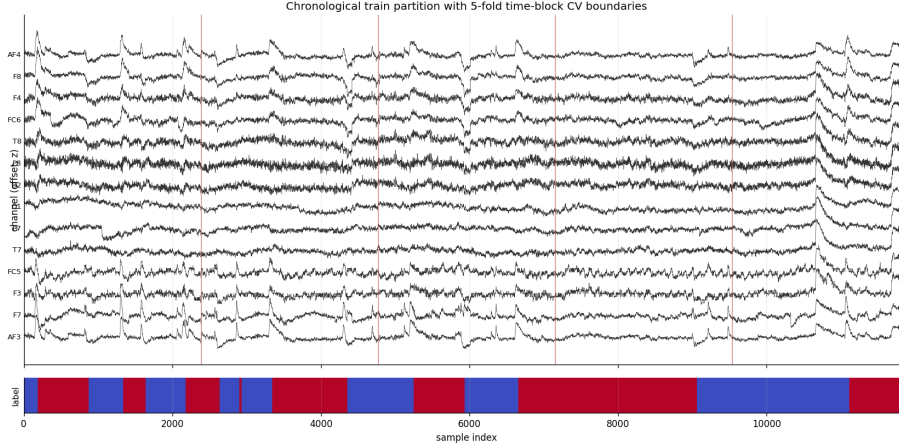


Figure 4: Per-fold sample assignments under the three CV schemes. Shuffled (top) scatters each fold across the whole recording, so temporal neighbours of a test sample are in the training folds. Naive blocked (middle) uses 5 long contiguous chunks with very uneven class balance. Stratified blocked (bottom) uses 100 short contiguous segments redistributed so each fold is class-balanced while neighbours stay together.

of fold i and the other four blocks are its training set. Because the permutation is uniform, each test fold is a random 20% subset spread across the whole recording—so the immediate temporal neighbours of almost every test sample land in the training folds. This is the standard “`KFold(shuffle=True)`” protocol, and the leakiest possible scheme on a continuous time series.

Scheme B — naive blocked 5-fold. We take the time-ordered index array *without* shuffling and cut it into 5 contiguous chunks. Fold i ’s test set is chunk i (a single contiguous span of time); its training set is the remaining time. This eliminates lag- ≤ 1 leakage but introduces a new problem: because the label runs are long and unevenly distributed, individual chunks can be near class-pure (e.g. one chunk at $\sim 20\%$ class-0 vs. the overall $\sim 54\%$). Per-fold accuracy variance therefore explodes and the mean can fall *below* the majority baseline. We include it to make the failure mode visible—it is *not* a scheme we recommend.

Scheme C — stratified blocked 5-fold (honest). This is the protocol we recommend. We cut the time-ordered training partition into $M = 100$ short contiguous segments of ~ 119 samples (~ 0.93 s) each, preserving within-segment temporal continuity. We label each segment by its class-1 proportion, sort the segments by that proportion, and walk the sorted list in chunks of 5; within each chunk the five segments are assigned to the five folds by a seed-fixed random permutation. The effect is that each fold receives one segment from each quintile of the class-1-proportion distribution, so every fold’s class balance is within ~ 3 pp of the overall $\sim 54\%$ class-0 balance. This keeps the temporal-isolation guarantee of blocked CV (neighbours stay together inside a segment) *and* restores the class-balance guarantee of stratified CV. Figure 4 visualises the per-fold sample assignments under all three schemes.

2.6 Comparison models (sanity checks)

To test whether the leakage gap is KNN-specific or a generic property of this dataset, we run the same model-selection idea on three other classifiers from the palette. **LDA** has no tunable hyperparameter—one accuracy per scheme. **PCA**→**LDA** sweeps $n_{\text{comp}} \in \{1, 2, 3, 5, 7, 10\}$ and selects the n maximising mean stratified-blocked CV accuracy. **PCR-as-classifier** (ordinary least squares on the first n principal components, thresholded at 0.5) sweeps the same grid. These are supporting evidence, not the primary result.

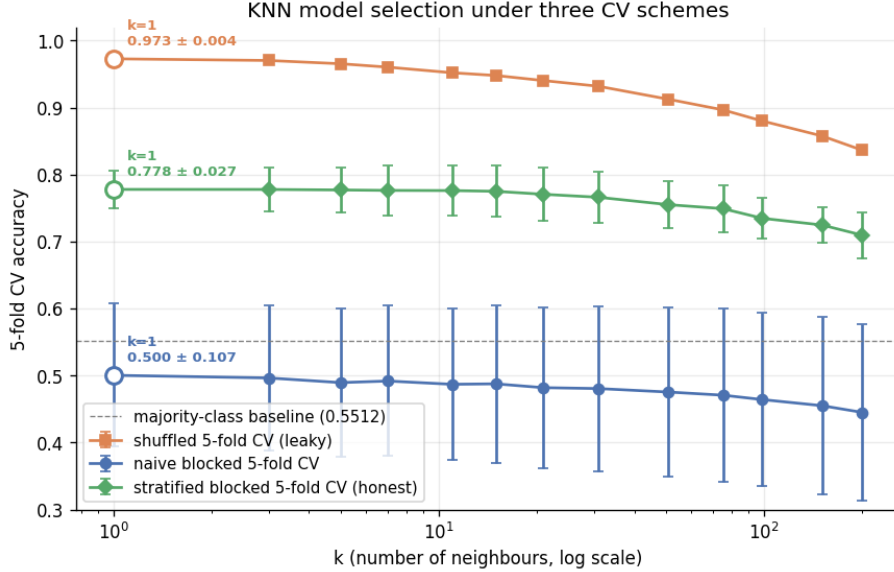


Figure 5: KNN model selection under three CV schemes (5-fold each; error bars are ± 1 SD across folds). Orange squares: shuffled CV (leaky baseline). Blue circles: naive blocked CV. Green diamonds: stratified blocked CV (honest). The horizontal dashed line is the majority-class baseline (0.5512). The open marker on each curve highlights the argmax-accuracy k and the corresponding accuracy.

Table 1: Per-scheme selected k and accuracy from the KNN k -sweep. All three schemes select $k = 1$; the reported accuracies span 47 percentage points.

CV scheme	Selected k	Mean accuracy $\pm$ SD
Shuffled (leaky baseline)	1	0.9728 ± 0.0037 ($\approx 97.3\%$)
Naive blocked	1	0.5004 ± 0.1068 ($\approx 50.0\%$)
Stratified blocked (honest)	1	0.7778 ± 0.0274 ($\approx 77.8\%$)

2.7 Metrics

We report **accuracy** $= (\text{TP} + \text{TN})/N$ as the headline metric for both CV and holdout, and **sensitivity** (recall for closed= 1) and **specificity** (recall for open= 0) in the holdout confusion matrices. The chronological holdout is heavily class-imbalanced ($\sim 91\%$ class-0, because the final segment of the recording is a long eyes-open run), so we treat the per-fold CV numbers, not the holdout accuracy, as the meaningful summary.

3 Results

3.1 Model selection: KNN under three CV schemes

Figure 5 is the headline figure. It plots KNN mean accuracy $\pm$ standard deviation as a function of k (log scale) under all three CV schemes on the same axes.

Three observations are immediate. The shuffled curve sits near 0.97 across all k , decaying roughly linearly on the log axis from 0.973 at $k = 1$ to ~ 0.84 at $k = 201$ —“nearly perfect” everywhere. The naive blocked curve sits near 0.50 across all k , at or below the majority baseline, with very large per-fold standard deviation (~ 10 – 13 pp); this is fold construction failing, not the model failing (§4.3). The stratified blocked curve peaks at 0.778 at $k = 1$ and decays gracefully to ~ 0.71 by $k = 201$, with a much smaller standard deviation (~ 2.7 – 4.0 pp).

Applying the argmax-accuracy rule to each scheme yields Table 1.

Table 2: Per-model leakage gaps (shuffled CV minus stratified-blocked CV) at the selected hyperparameter. KNN’s gap is the outlier: $\sim 4\text{--}8\times$ larger than any other classifier in the palette.

Model	Selected hyperparam.	Shuffled CV	Strat. blocked CV	Leakage gap
LDA	— (no sweep)	0.6471 ± 0.0063	0.5911 ± 0.0474	$+5.6\text{ pp}$
PCA→LDA	$n = 3$	0.5574 ± 0.0147	0.5339 ± 0.0438	$+2.4\text{ pp}$
PCR (classifier)	$n = 2$	0.5528 ± 0.0087	0.5250 ± 0.0176	$+2.8\text{ pp}$
KNN	$k = 1$	0.9728 ± 0.0037	0.7778 ± 0.0274	$+19.5\text{ pp}$

The remarkable result is that **all three schemes select the same hyperparameter** ($k = 1$): the model-selection procedure is robust, but the *answer* it reports is not. The selected accuracy ranges from 50.0% (naive blocked) to 97.3% (shuffled)—a 47 pp spread—despite the procedure, the model, the data, and the selected k being identical.

3.2 Complexity, flexibility, and overfitting

For KNN, model flexibility is inversely related to k : $k = 1$ is the most flexible (a jagged, high-variance decision boundary that can fit any training point), and large k is more rigid (a smooth, high-bias boundary). The shape of the curves in Figure 5 is itself diagnostic of overfitting-via-leakage. Under *honest* stratified blocked CV, accuracy *decreases* monotonically as k grows from 1 to 201 ($0.778 \rightarrow 0.710$): the flexible model genuinely generalises best, with no sign of the usual “too-flexible” overfitting dip, because the signal it exploits (short-range temporal smoothness within a segment) is real and local. Under *shuffled* CV the curve is pathologically flat-and-high (~ 0.97 everywhere): the model is memorising the temporal twin of each test sample regardless of k . That flat-near-1 curve is the fingerprint of leakage-driven overfitting—the estimator looks like it cannot overfit because the “test” samples are duplicates of training samples. The honest curve is the one that behaves like a real bias–variance trade-off.

3.3 Leakage-gap analysis

The methodological centrepiece is the **leakage gap**: the difference between an estimator’s leaky shuffled-CV accuracy and its honest stratified-blocked-CV accuracy at the same selected hyperparameter. For KNN at $k = 1$,

$$\text{leakage gap} = 0.9728 - 0.7778 = 0.1950 \quad (\approx 19.5\text{ pp}).$$

We attribute this gap to leakage of adjacent samples across the train/test fold boundary, not to model quality: the model is 1-NN in both cases and the training partition is identical—only the fold-assignment rule changes. Under shuffled CV the temporal neighbour of every test sample lands in a training fold with $\sim 80\%$ probability and carries the same label with $\sim 99.7\%$ probability; under stratified blocked CV that neighbour is reserved into the same segment as the test sample. The honest accuracy at $k = 1$, $77.8\% \pm 2.7\%$, is nonetheless $\sim 23\text{ pp}$ above the 55.12% baseline—the classifier is far from useless, just far less impressive than the shuffled number implies. **Both parts of our hypothesis are therefore supported.**

3.4 Supporting evidence: alternative classifiers

To rule out a generic explanation, we ran the same idea on three other classifiers (Table 2, Figure 6).

The pattern is exactly what we expect if the gap is caused by per-sample autocorrelation. LDA’s decision rule is a single global linear projection of the 14-D space onto a 1-D axis; it cannot exploit per-sample similarity, so its gap is small (5.6 pp). PCA→LDA and PCR both project to a low-dimensional subspace before deciding, destroying most of the high-frequency sample-to-sample similarity KNN relies on; their gaps are smaller still (2.4 and 2.8 pp). KNN at $k = 1$ is the only classifier whose prediction depends on the identity of the single nearest training sample—and its gap is $4\text{--}8\times$ larger than any other.

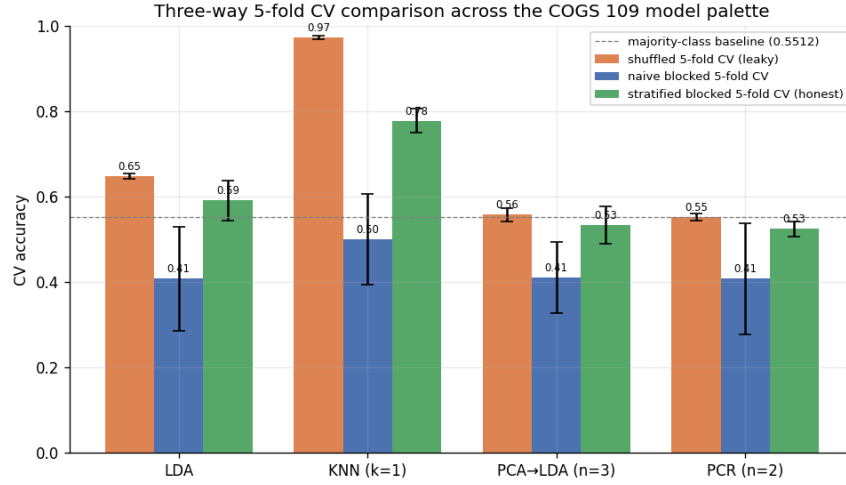


Figure 6: Three-way 5-fold CV accuracy across the four palette classifiers. Orange: shuffled; blue: naive blocked; green: stratified blocked. The KNN bar group has by far the largest shuffled-to-stratified-blocked gap, consistent with KNN being uniquely vulnerable to lag-1 autocorrelation.

3.5 Model estimation: final parameters and accuracy

Final model and “parameters”. The selected model is KNN with $k = 1$ on the z -scored 14-channel features. KNN is non-parametric, so the “fitted parameters” are not coefficients but (i) the selected hyperparameter $k = 1$ and (ii) the stored training set itself together with the per-channel z -scoring constants (μ_c, σ_c) estimated on the full training partition (persisted in `data/processed/scaler.json`). For a concrete parametric contrast, the LDA comparison model *does* have estimable coefficients: its 14-channel discriminant weights (estimated on the full training set) are reported in `figures/10_lda_coefficients.png`, with the occipital/parietal channels (O1, O2, P7, P8) carrying the largest magnitude—consistent with eye-closure altering posterior alpha rhythms.

Final accuracy. The honest cross-validated estimate of the selected model ($k = 1$, stratified blocked CV) is **77.8% $\pm$ 2.7%**, versus the 55.12% majority baseline. As a confirmatory out-of-sample check, Figure 7 shows per-model confusion matrices on the 20% chronological holdout. The holdout is $\sim$ 91% class-0, so its accuracy is dominated by specificity and we do not over-interpret it; we did not rebalance it because doing so would itself introduce leakage. The CV numbers above are the headline estimate.

4 Conclusions and Discussion

4.1 What we conclude about the hypothesis

Both parts of our hypothesis hold. (i) The honest KNN accuracy (77.8%) beats the majority baseline (55.12%) by $\sim$ 23 pp, so raw 14-channel voltages do carry a genuine, generalisable eye-state signal. (ii) The shuffled-CV estimate (97.3%) is inflated by 19.5 pp relative to the honest estimate, and that inflation is leakage attributable purely to the CV scheme: the model, the selected k , and the training data are identical across schemes.

Shuffled CV says we have a 97% classifier. Honest stratified blocked CV says we have a 78% classifier. The difference is leakage, not signal.

4.2 Why KNN is uniquely vulnerable

A 1-NN classifier predicts the class of the single nearest training neighbour. Because the lag-1 channel correlation is ≈ 0.997 , the nearest neighbour in feature space is almost always the temporal neighbour.

Confusion matrices on chronological holdout (n=2,995)

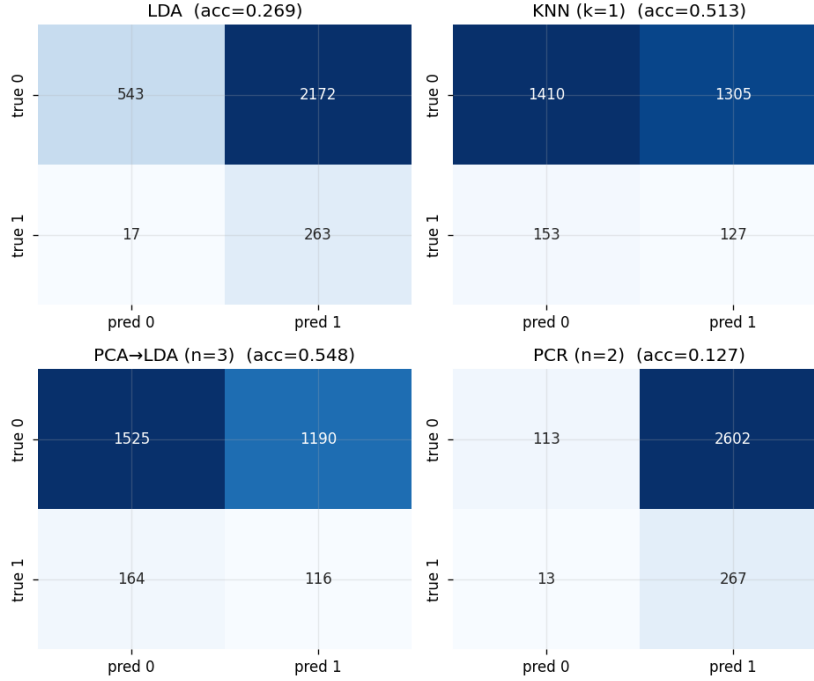


Figure 7: Per-model confusion matrices on the 20% chronological holdout test set (rows: true; columns: predicted). The holdout is dominated by eyes-open samples ($\sim 91\%$ class-0), so accuracy here is mostly driven by specificity. The number in each panel is that model’s holdout accuracy.

Under shuffled CV that neighbour is in a training fold $\sim 80\%$ of the time and shares the test label $\sim 99.7\%$ of the time, so 1-NN trivially exceeds 95% without learning any voltage-to-state rule. Under stratified blocked CV the temporal neighbour is reserved into the test sample’s own segment, forcing KNN onto the next-nearest non-adjacent sample, and accuracy drops to 77.8%. Models that average over many samples (LDA, PCA→LDA, PCR) cannot exploit this and show only 2–6 *pp* gaps.

4.3 Why naive blocked CV underperforms the baseline

The five contiguous chunks are not class-balanced: with only 23 label transitions in the recording, several chunks straddle few or zero transitions, and per-chunk class proportions stray far from the global 55/45 split. A classifier trained on four mostly-open chunks then predicts “open” on a held-out mostly-closed chunk, scoring *below* the majority baseline. This is a fold-stratification failure, not a model failure; stratifying contiguous segments (Scheme C) fixes it, which is why stratified blocked CV recovers ~ 28 *pp* over naive blocked.

4.4 Implications and future work

A non-trivial fraction of tutorials and student projects on this dataset report 95–98% KNN accuracy under shuffled k -fold. Our analysis suggests those numbers are a CV-protocol artefact: the honest accuracy is 77.8% and the extra ~ 20 *pp* is adjacent-sample leakage. This echoes broader cautions about i.i.d. CV on autocorrelated signals (Bergmeir & Benítez, 2012). The practical recommendation is to use blocked, class-stratified CV (or an explicit out-of-time holdout with a seam gap) whenever samples are temporally autocorrelated. Limitations of our study: it is a single subject on consumer hardware over ~ 117 s, and we use per-sample voltages with no windowed features. Future work should (i) add windowed features (rolling means/variances, lagged values), which would give KNN more meaningful neighbours and likely shrink the leakage gap; (ii) test cross-subject generalisation on a multi-subject

corpus; and (iii) extend the k sweep to characterise the leakage curve more finely. The project is, above all, a worked example of how a methodologically sound model-selection procedure can still be misled by an upstream choice—the CV scheme—that has nothing to do with the model itself.

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